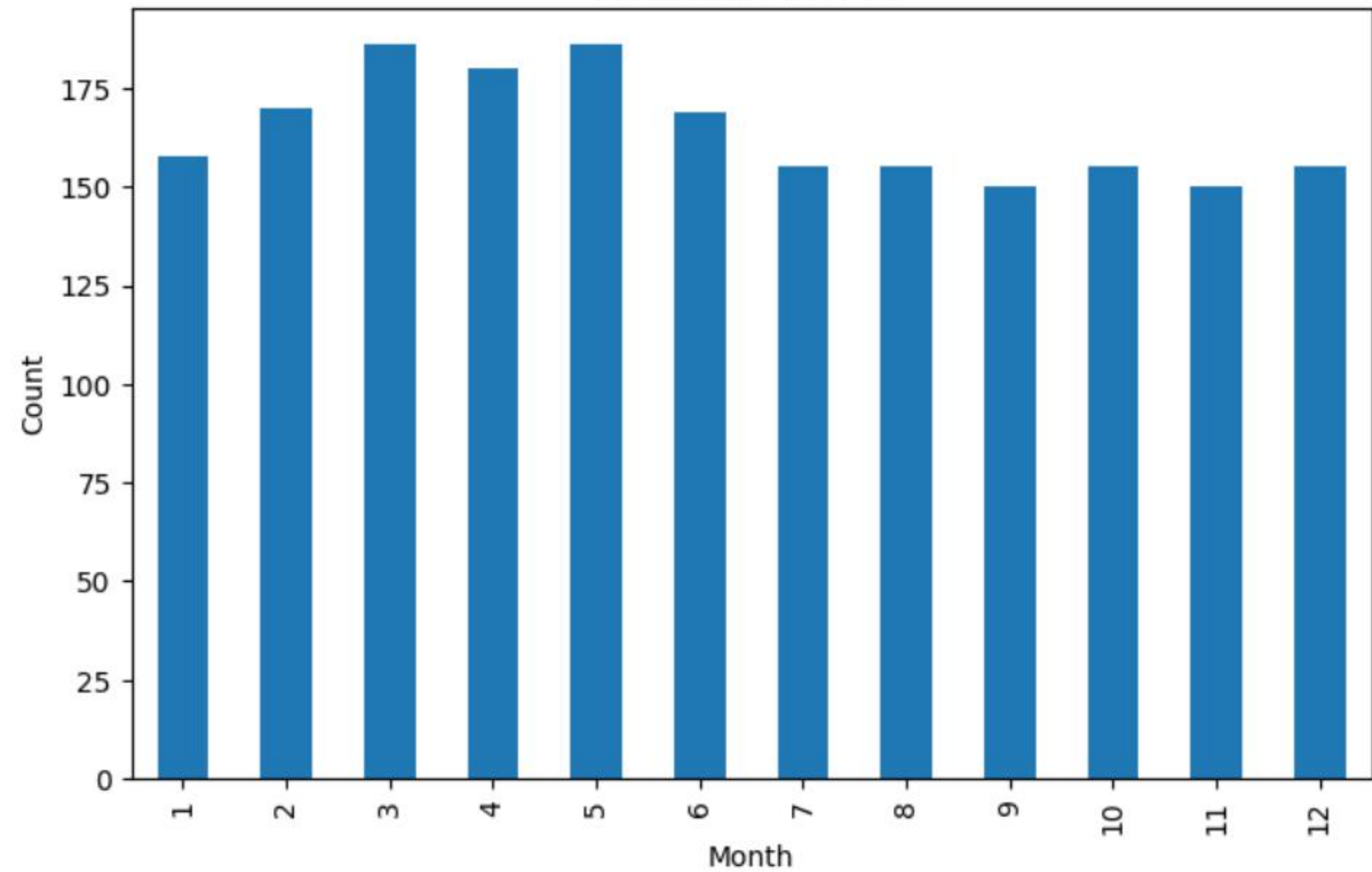
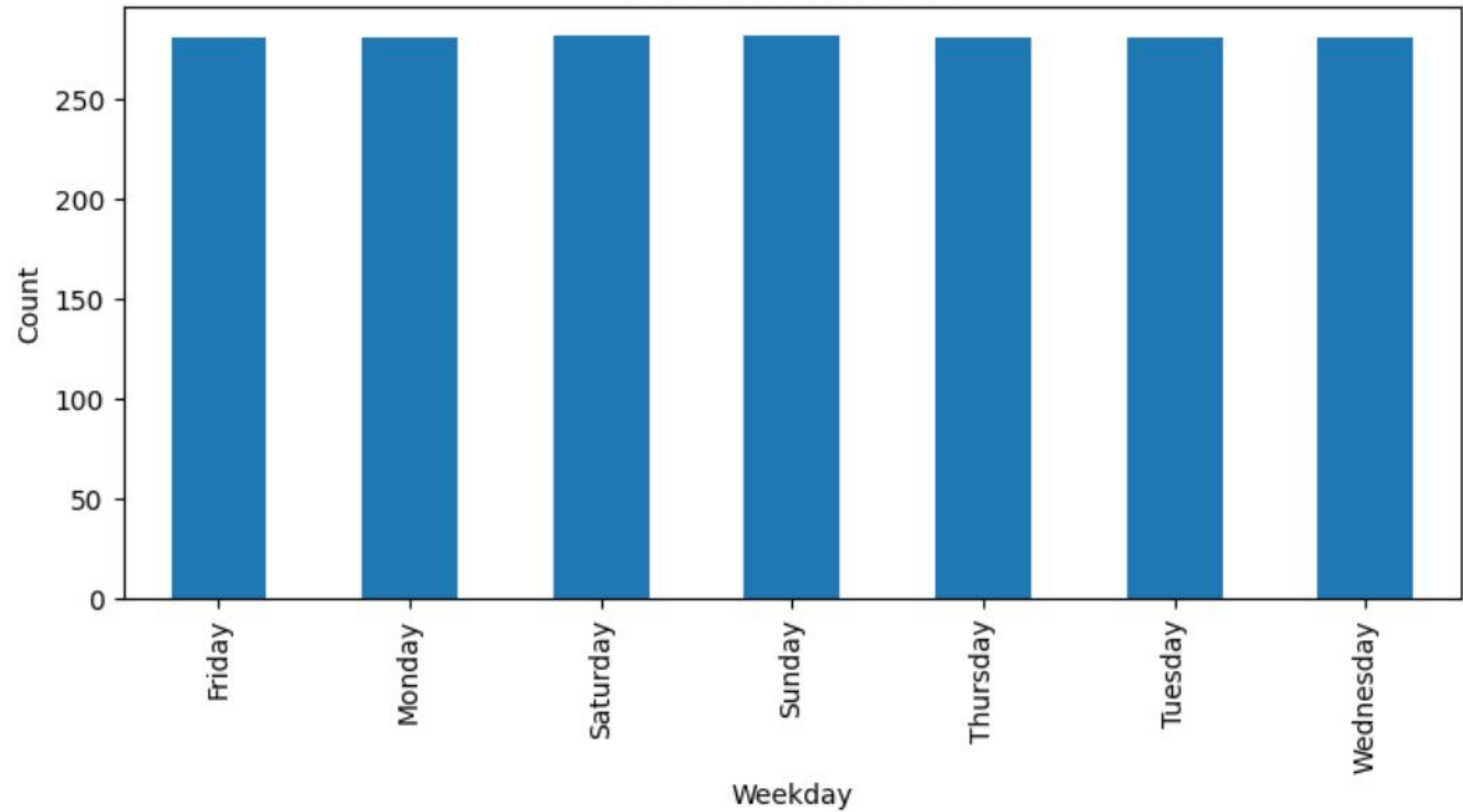


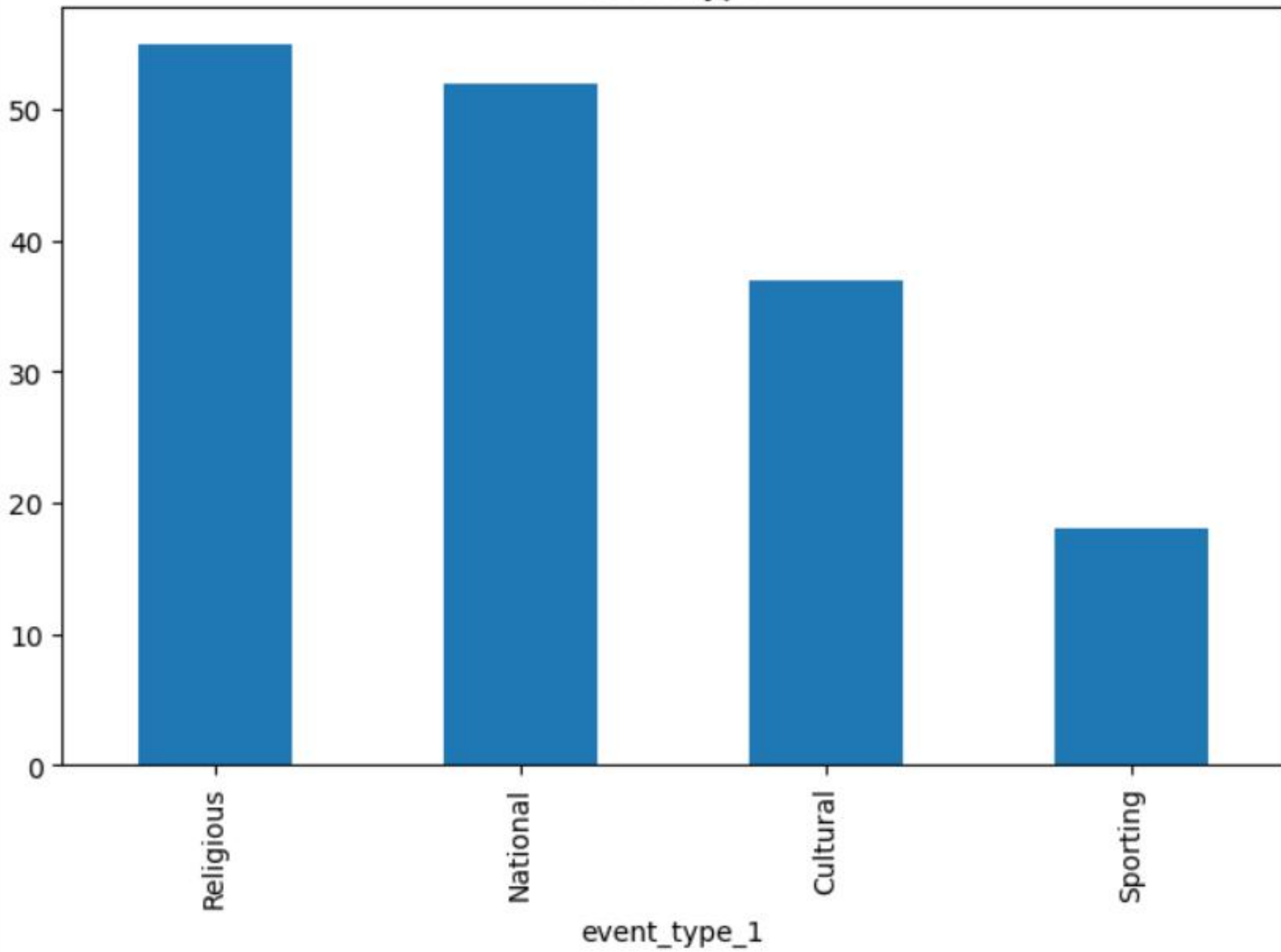
Month Distribution



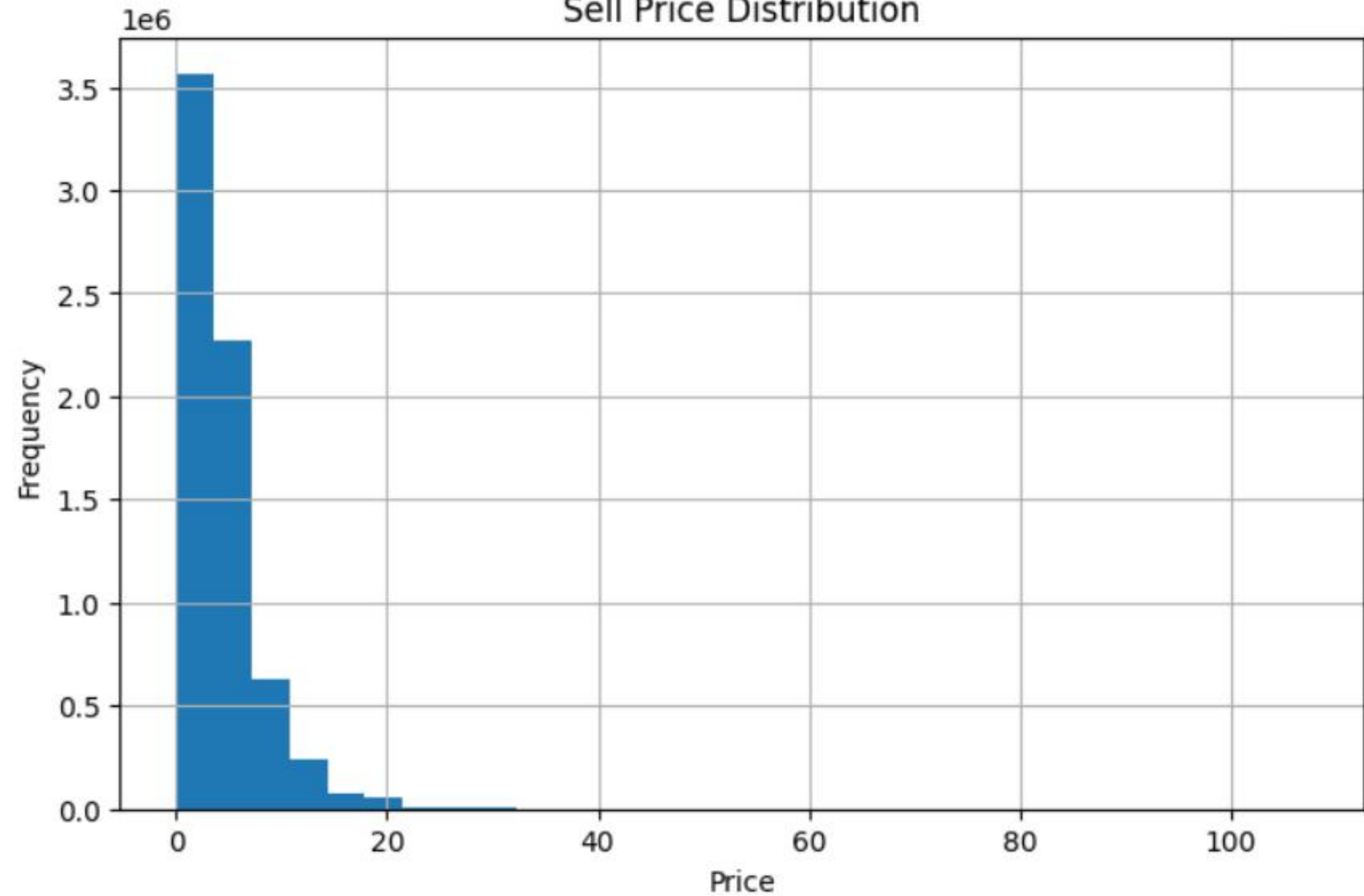
Weekday Distribution



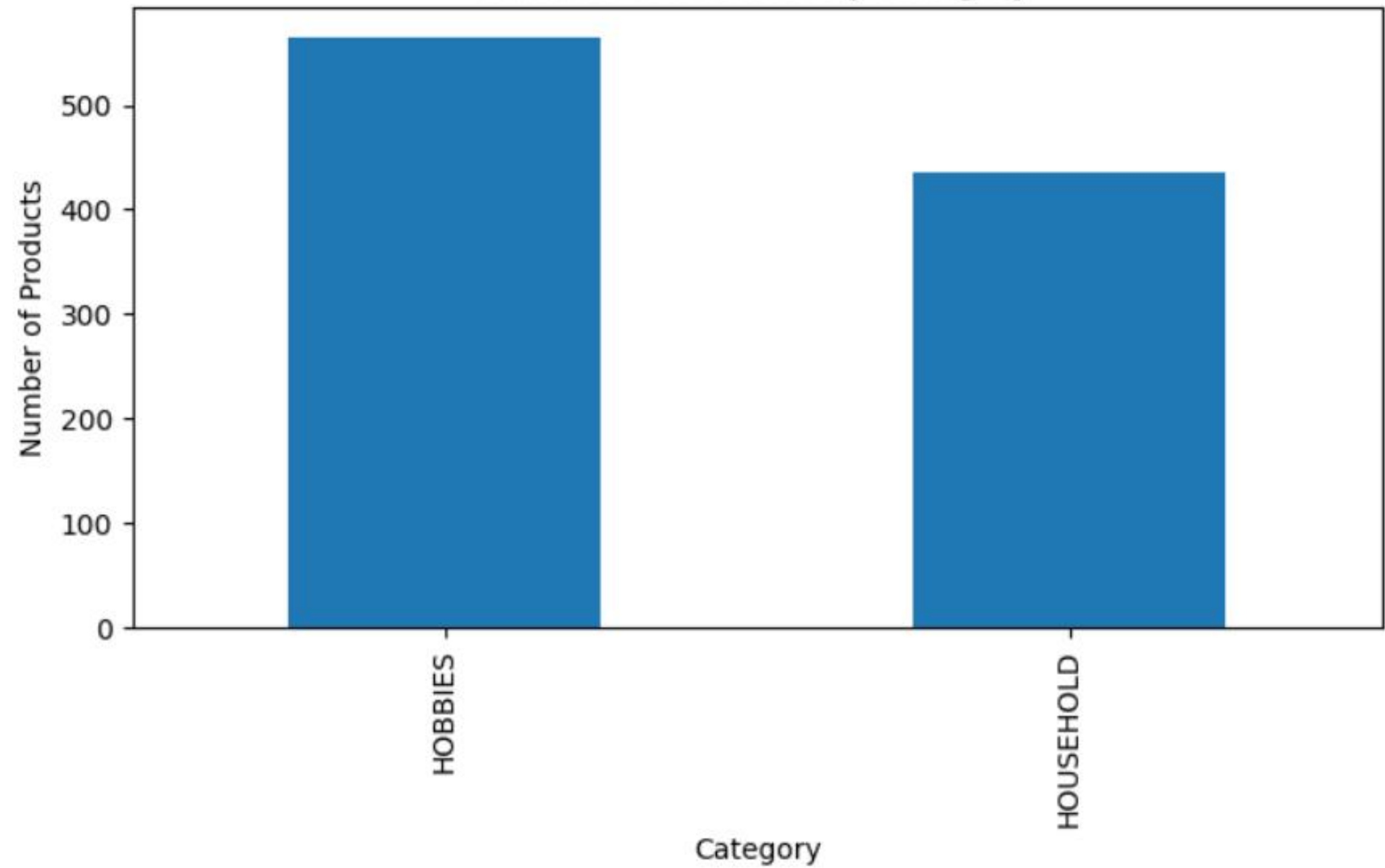
Event Type



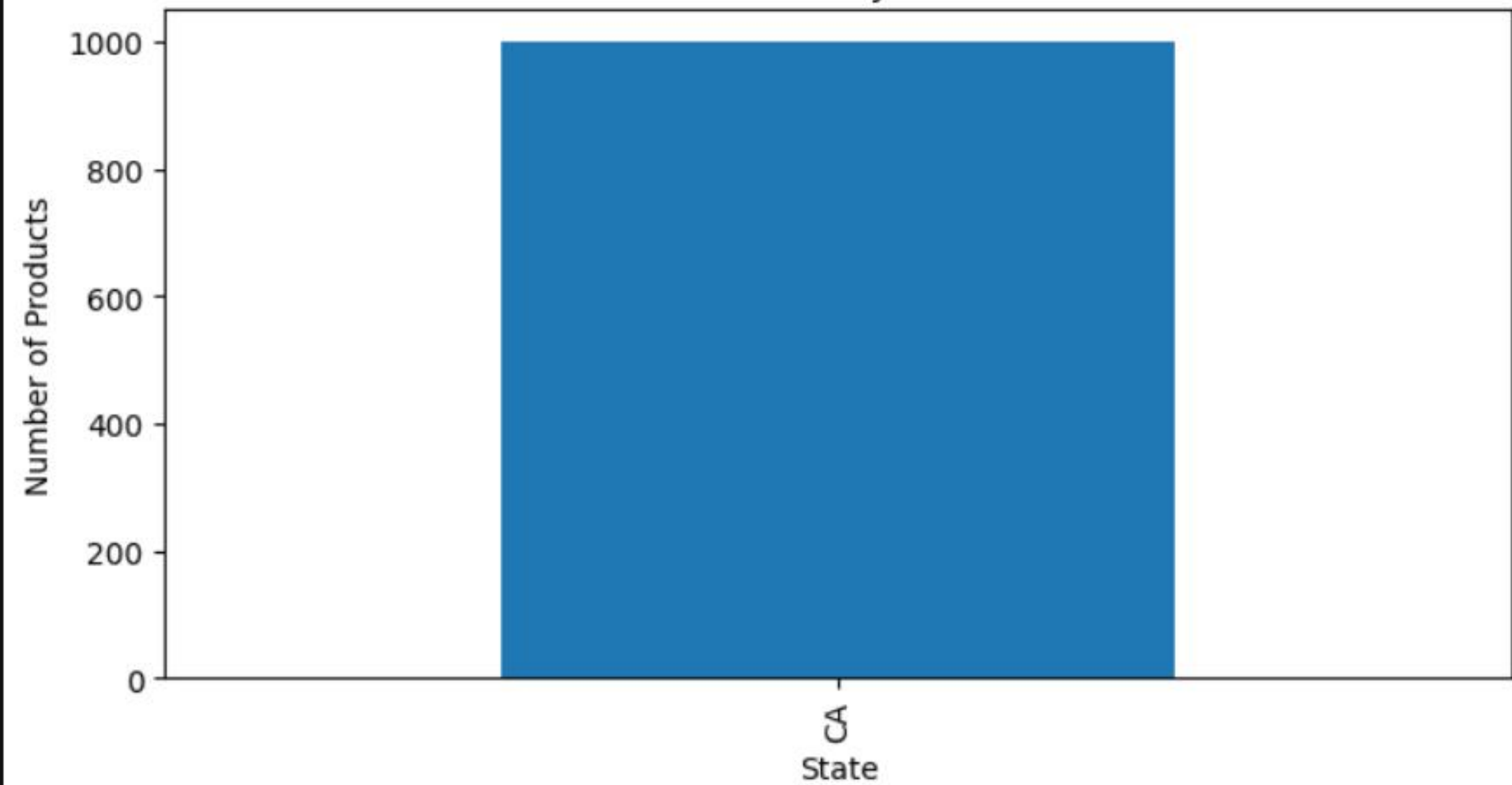
Sell Price Distribution



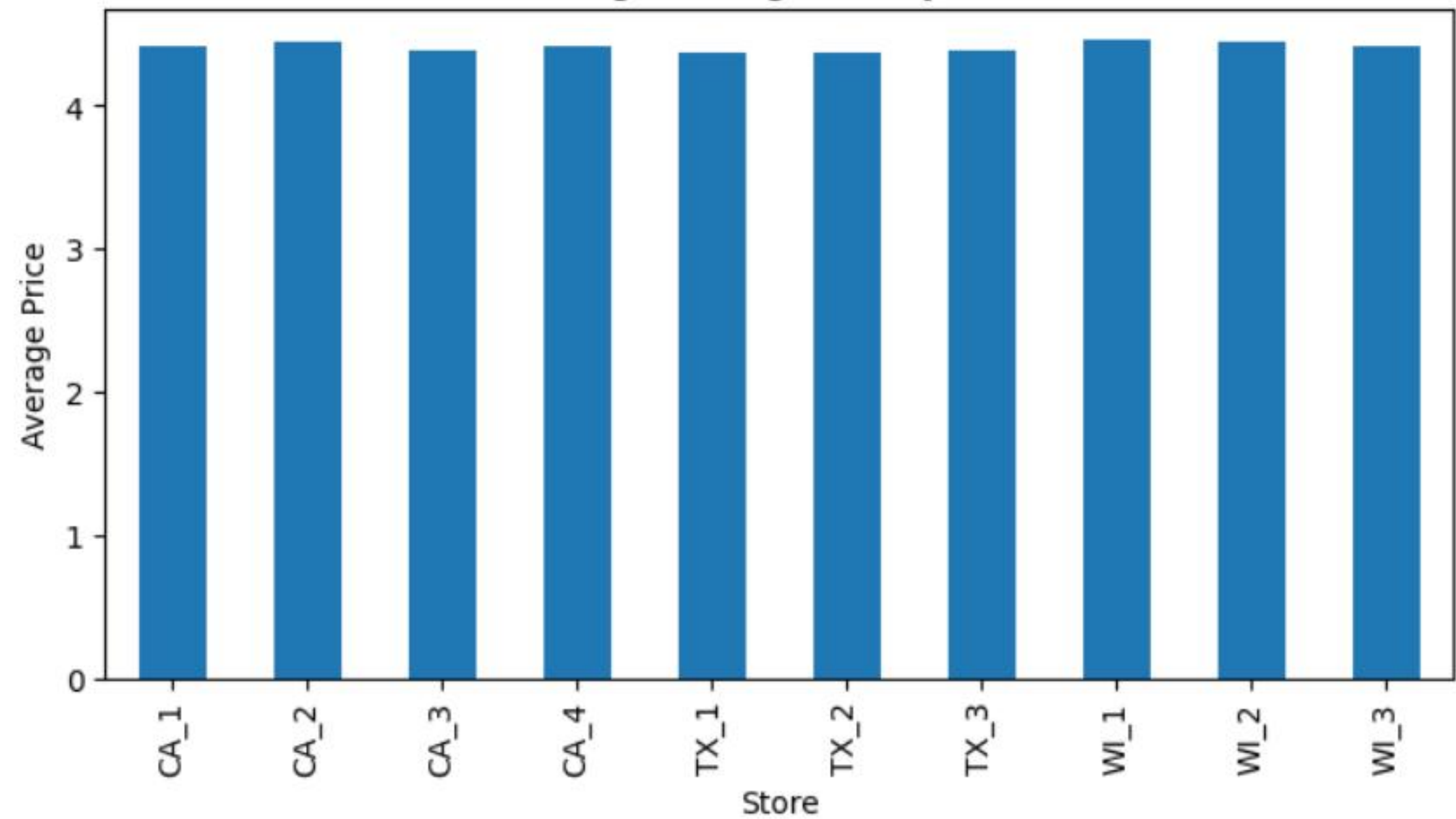
Number of Products by Category



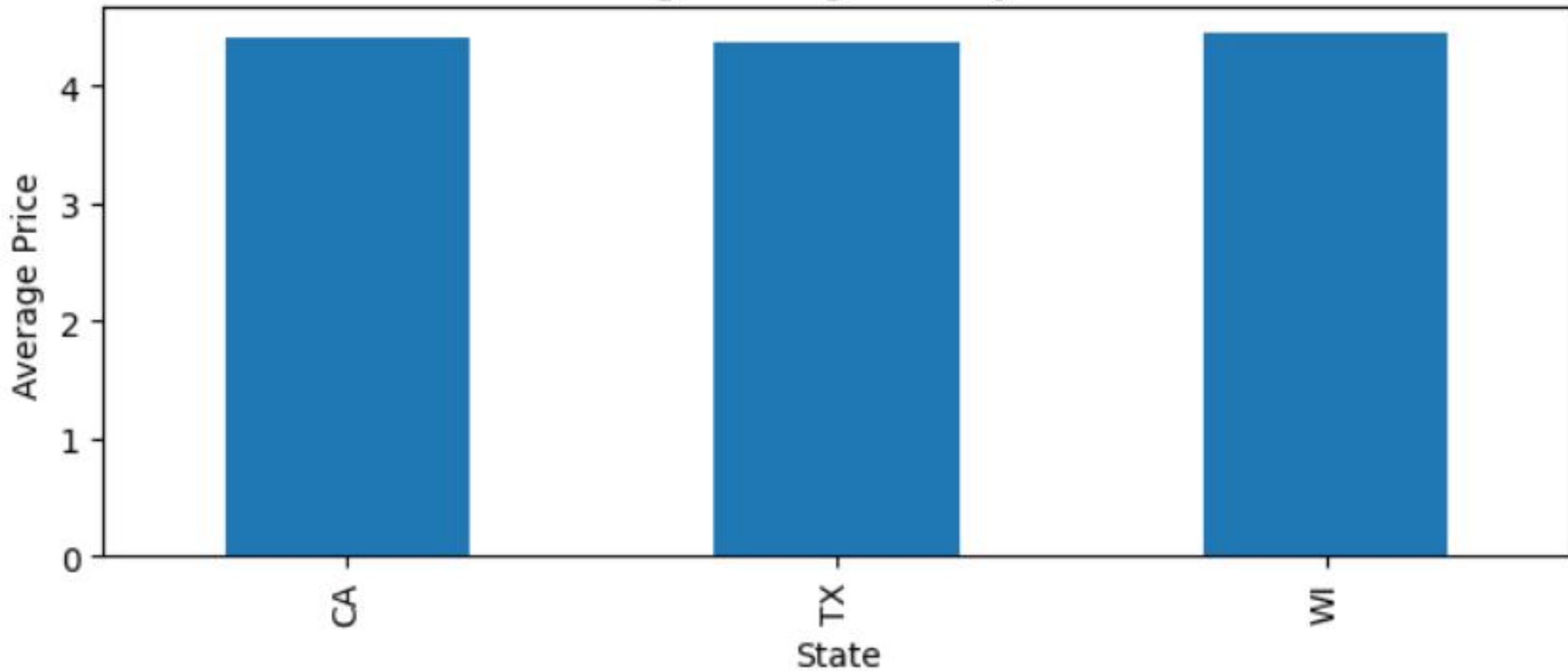
Products by State



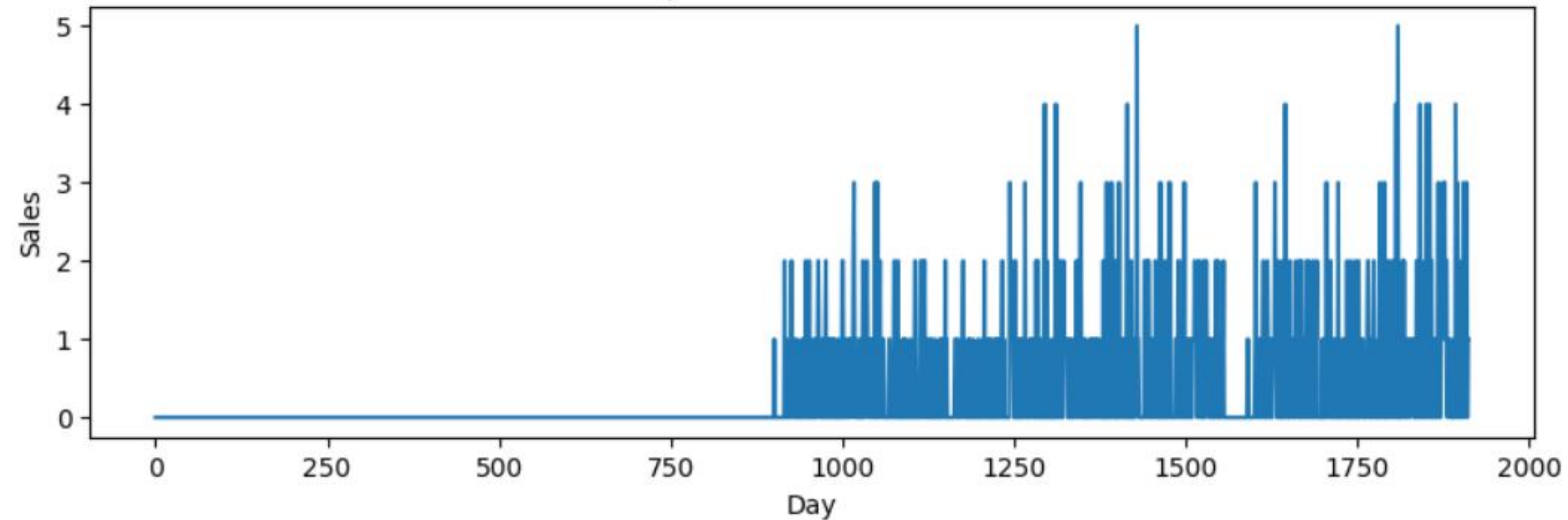
Average Selling Price by Store



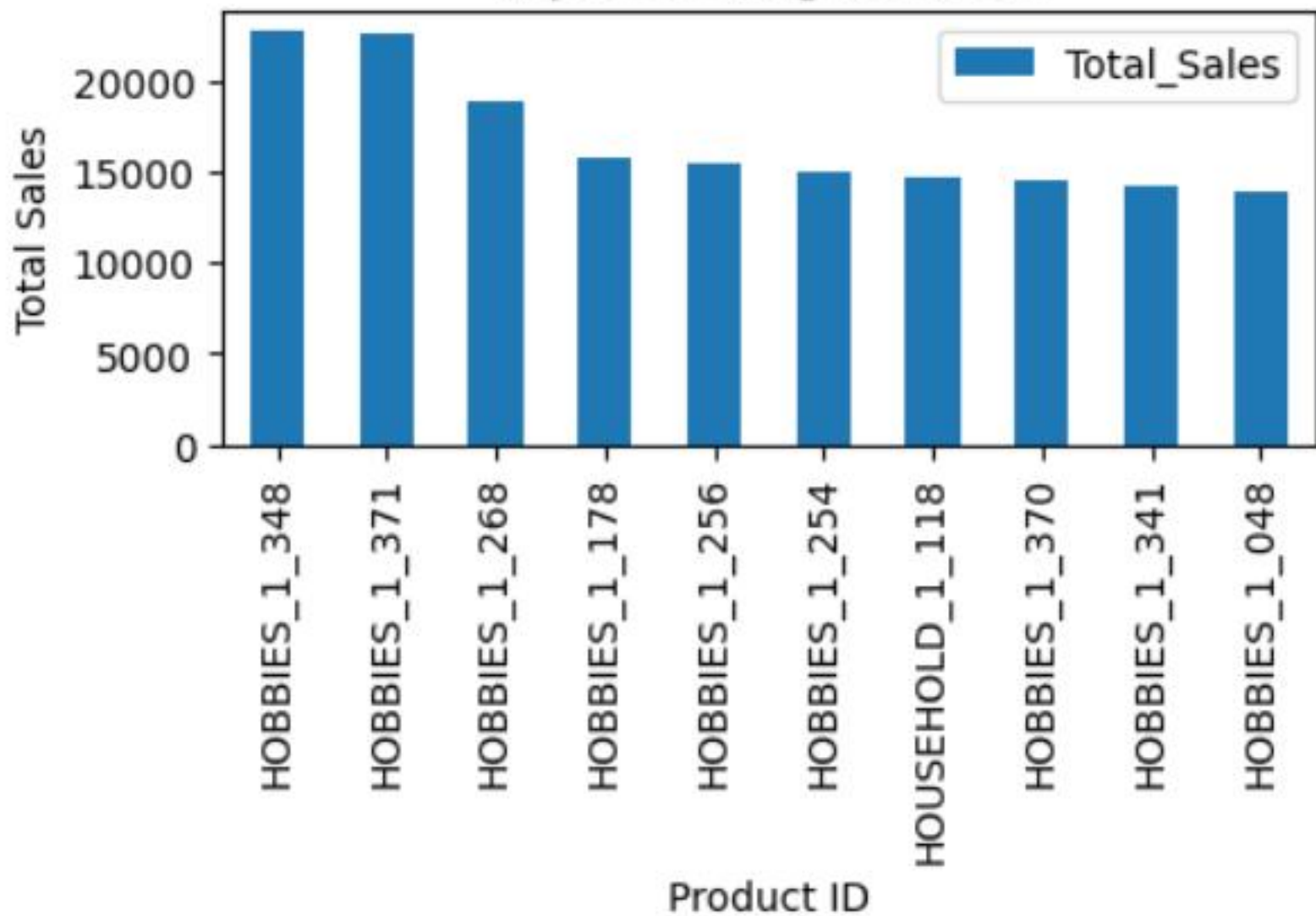
Average Selling Price by State



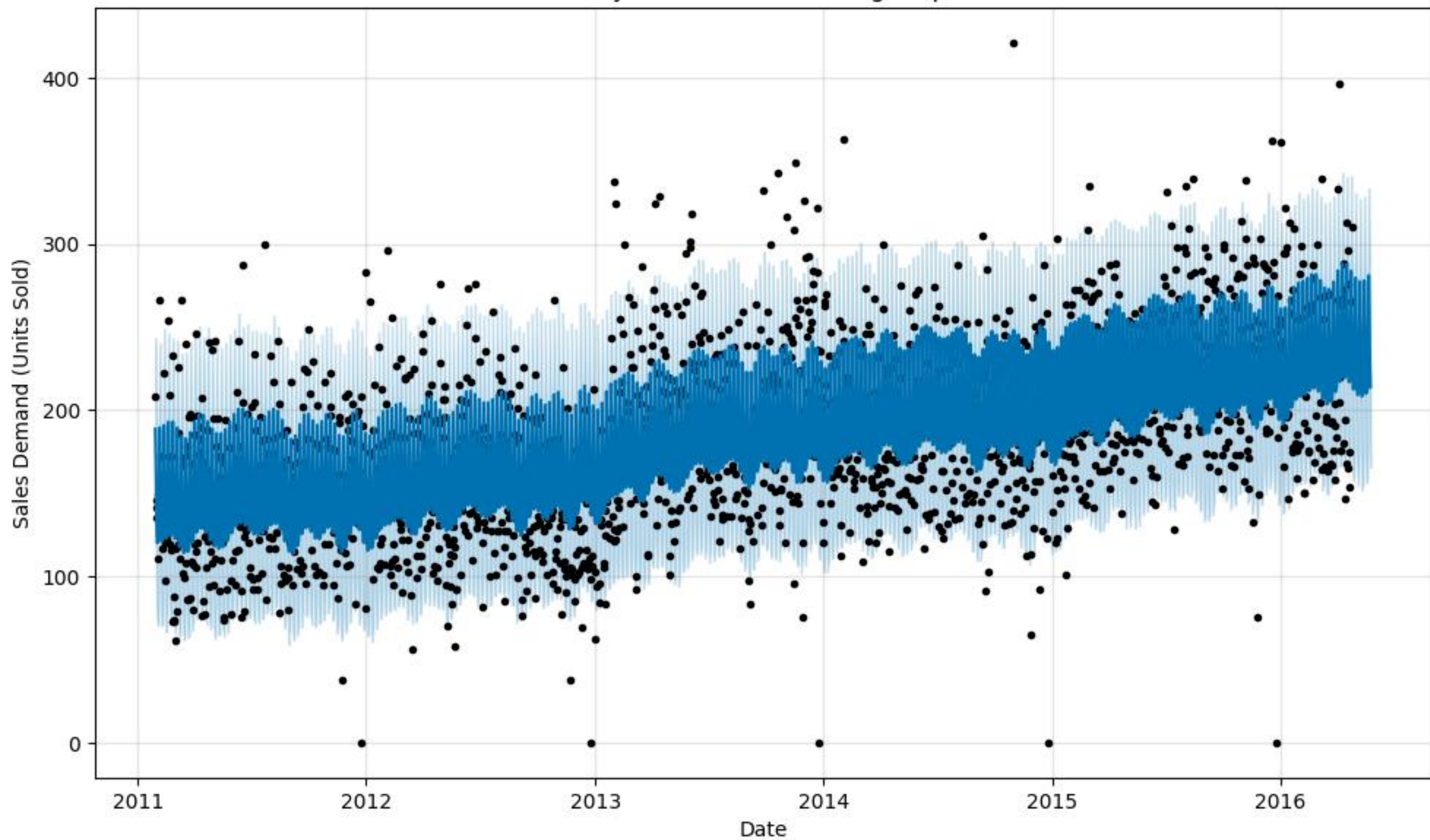
Daily Sales of First Product



Top 10 Selling Products

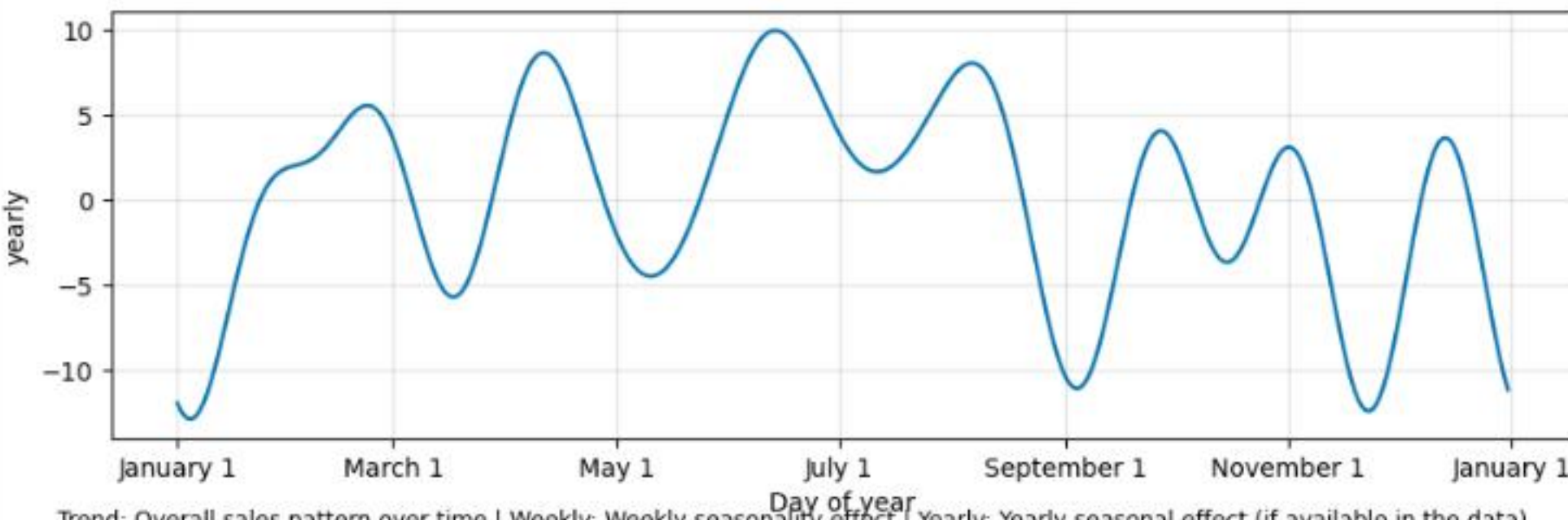
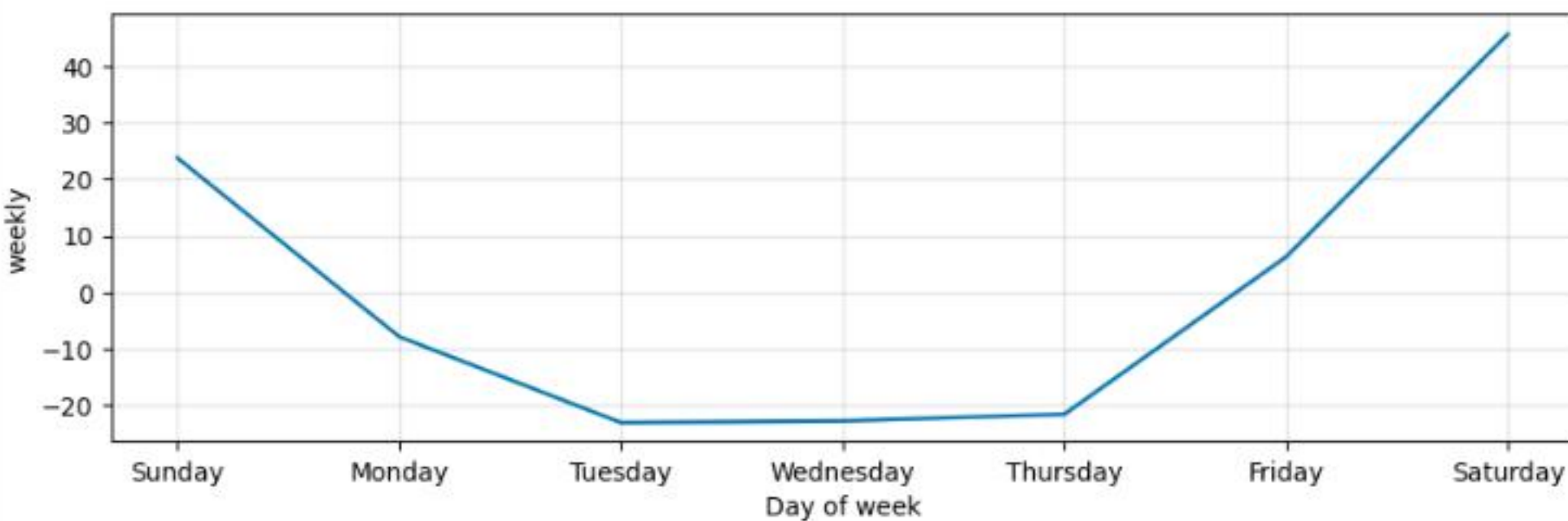
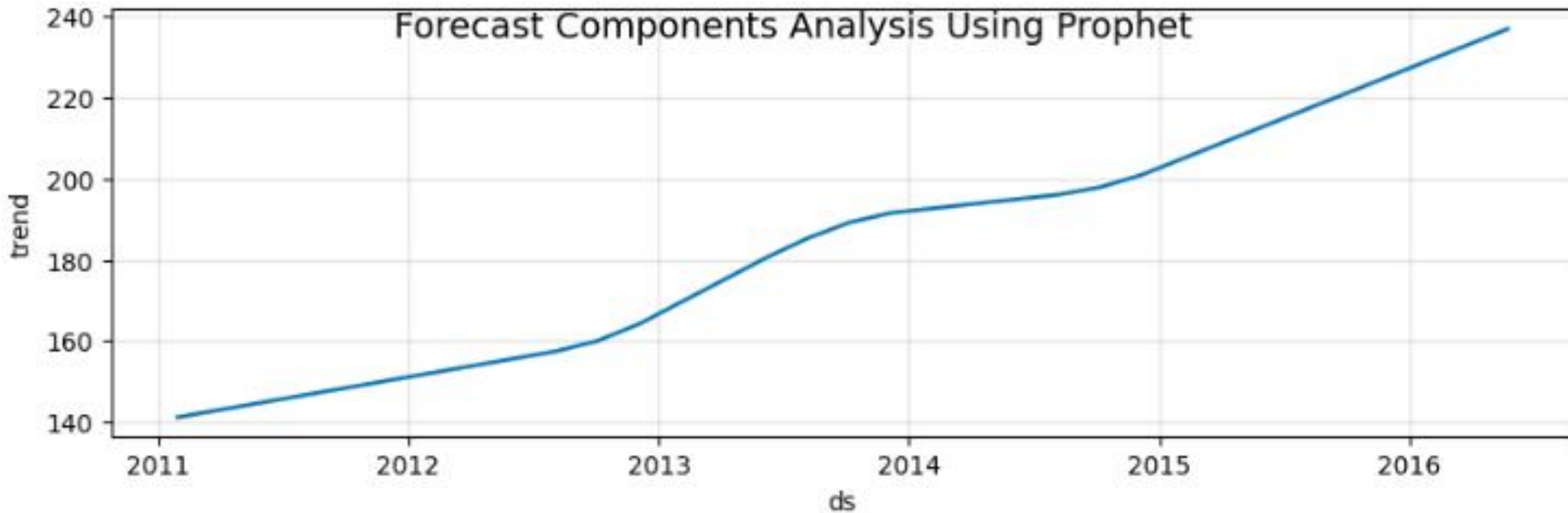


30-Day Sales Forecast Using Prophet



Black dots: Actual historical sales | Blue line: Forecasted sales (yhat) | Light blue shaded area: Prediction interval (yhat_lower to yhat_upper)

Forecast Components Analysis Using Prophet



Trend: Overall sales pattern over time | Weekly: Weekly seasonality effect | Yearly: Yearly seasonal effect (if available in the data)

30-Day Sales Forecast

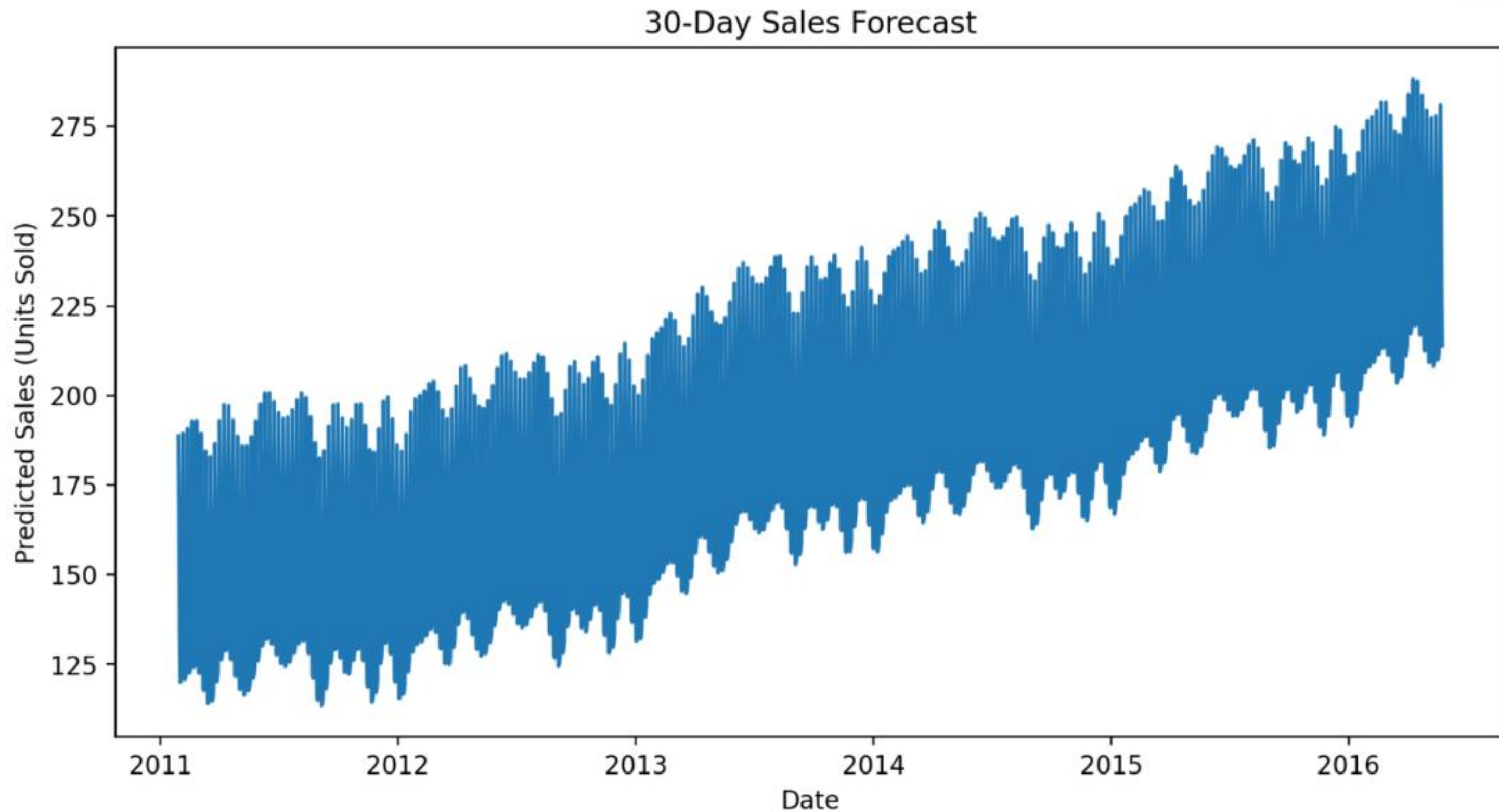


Figure 1: 30-Day sales forecast generated using the Prophet model. X-axis: Date | Y-axis: Predicted Sales (Units Sold)

Sales Summary

Total Forecasted Sales: 356836.73

Average Predicted Sales: 183.65

Maximum Predicted Sales: 288.12

Minimum Predicted Sales: 113.53

Dataset Information

Dataset: M5 Forecasting Dataset

Forecast Records: 1943

Forecast Horizon: 30 Days

Domain: Retail Sales Forecasting

Model Information

Model: Facebook Prophet

Algorithm Type: Time Series Forecasting

Input: Historical Retail Sales Data

Output: 30-Day Sales Forecast

Purpose: Forecast future product demand to support inventory planning and business decision-making.